Lecture 9: Network effects

Green vehicle adoption and which side of the market to subsidize

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Outline of today

1. Technology diffusion and network effects

2. Article for today: Li et al. (2017)

   Microfounding the supply side

   The demand side

   Transition dynamics

   Cost-effective EV policies

   Estimation and identification

   Counterfactual scenario

3. Conclusion

Technology diffusion and network effects

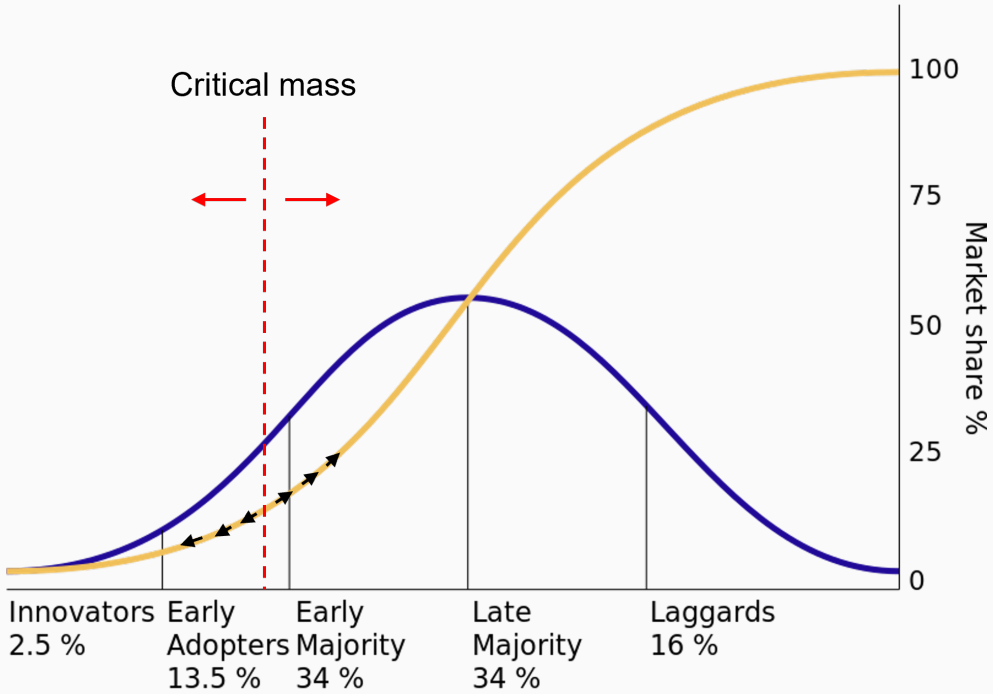
Technology diffusion and the green transition

- Transitioning to green energy requires the diffusion of new (green) technologies.
- We should care about
  1. How fast new technologies spread.
  2. How we optimally regulate the diffusion of new technologies.

Critical mass and technology diffusion

Rogers (1962): How and at what rate new technology spreads.

Critical mass: The point at which a technology's diffusion becomes self-reinforcing through positive feedback mechanisms.



Note: Orange line is the market share, and the blue line is the share of new adopters. Late majority share is incorrect.

Source: `https://en.wikipedia.org/wiki/Diffusion_of_innovations`

Technology adoption

Diffusion of innovations: Both the *creation* and *adoption* of technologies.

This course: (Indirect) Network effects in technology adoption

Only one channel of positive feedback mechanisms in technology diffusion, but many more might exist, e.g.

- Peer effects (direct network effects)
- Directed technical change
- Learning-by-doing
- Coordination errors

If some of these are market imperfections, Pigouvian taxes are not sufficient.

Why do network effects matter for the green transition?

- Green vehicle adoption is expected to be a key channel for abating emissions from transport in the future.
- An important question is whether the adoption of alternative vehicles depends on the infrastructure supplying the propellant.
- In the case of electric vehicles (EVs), are consumers willing to purchase an EV if the charging network is underdeveloped?
- Will firms invest in new charging stations if there are no EV users?

  Are EVs subject to *indirect network effects*?
- What are the policy implications in the presence of indirect network effects?

Direct vs Indirect Network Effects

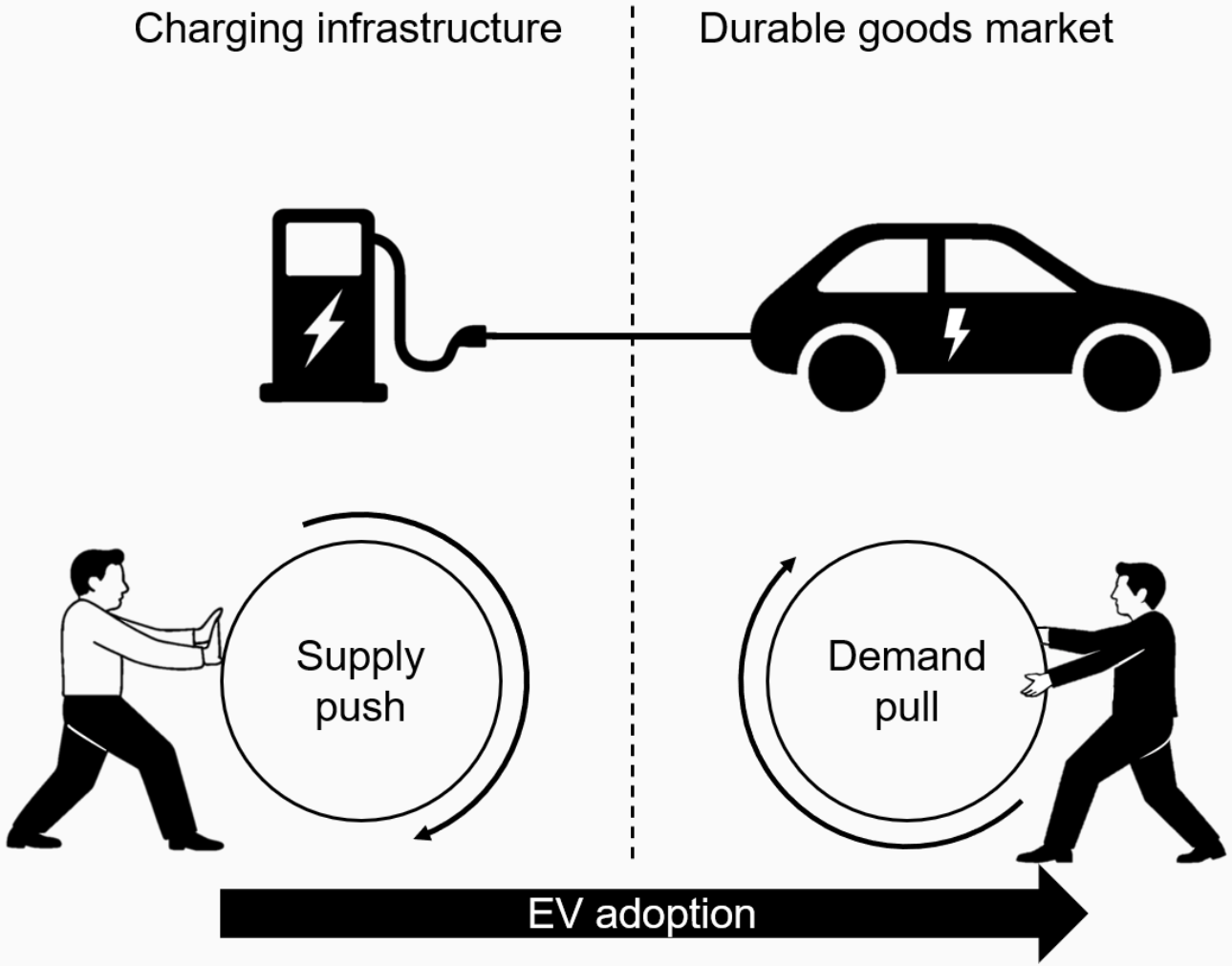
Positive network effect: If an agent's use of a good (i) benefits other users of a good and (ii) increases other users' incentives to use the good as well.

- *Direct:* Other agents' use of the good enters the utility of a consumer directly (e.g. Facebook). See e.g. Bollinger and Gillingham (2012).
- *Indirect:* Other agents' use of the good increases complementary services, making the good more valuable to the individual consumer (e.g. iPhones and Apple Store).

- Indirect network effects arise in *hardware/software systems*: consumers adopt the hardware if there is software, and developers produce software if there are enough users.
- The EV is the hardware and the charging network the software.

Supply push or demand pull strategies?

Are there arguments for supply push instruments or demand pull instruments?



Push vs. pull policies in a Danish context

The *Commission for the Green Transition of Passenger Cars*:

- Pigouvian taxes on car use.
- Rebates on EV investments.



Alternative:

- Not as difficult as the commission suggests due to indirect network effects.
- Subsidies for charging station deployment.



Ved at kombinere støtte til ladestandere med støtte til elbiler bliver omstillingen af bilparken billigere for samfundet, end Eldrup-kommissionen har beregnet.

Grøn omstilling af bilparken: Måske er missionen ikke så umulig

Undersøgelser fra Norge viser, at det er billigere at udbrede elbiler ved at give støtte til ladestandere end at øge støtten til selve bilen. Foto: Emil Helms/Ritzau Scanpix

PETER BIRCH SØRENSEN
JANEK BLIGAARD ESKILDSEN

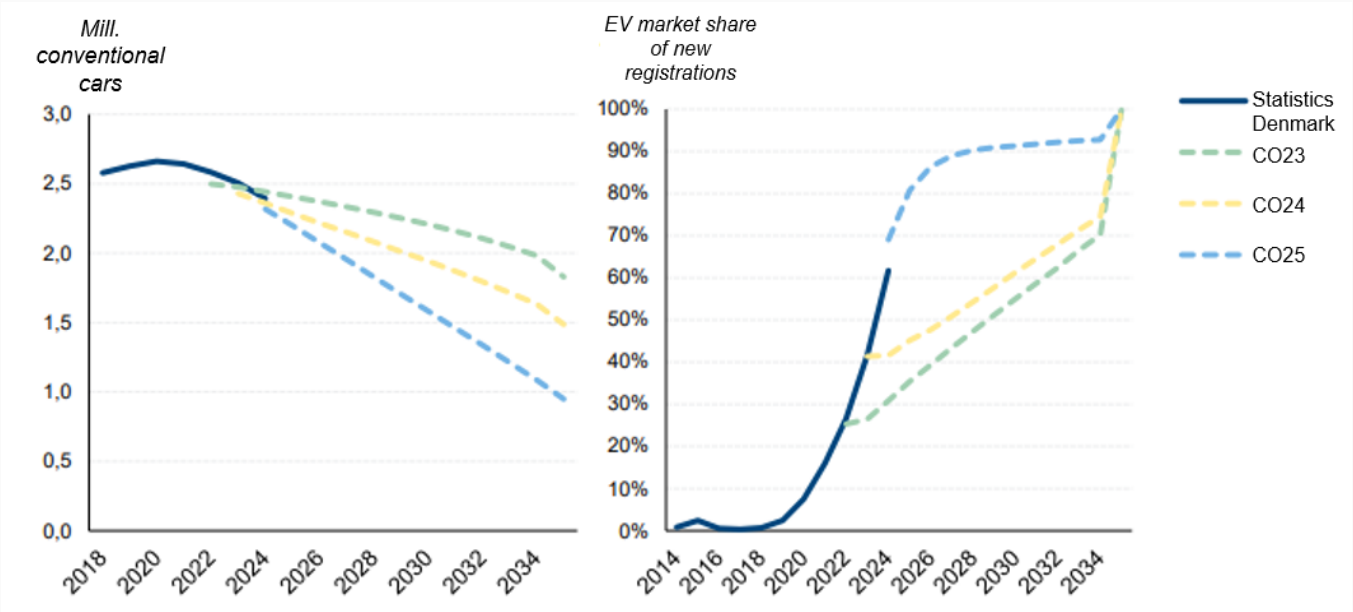
LYT TIL ARTIKLEN
GEM ARTIKLEN

KLIMA
17. SEPTEMBER 2020, 17.08

KOMMENTARER

The expected rate of EV adoption in Denmark

The Danish government has consistently underestimated the rate of EV adoption.



Note: 'CO' stands for Climate Outlook and is the government's official projection of emissions.
Source: Klimaraadet (2026, figure 1)

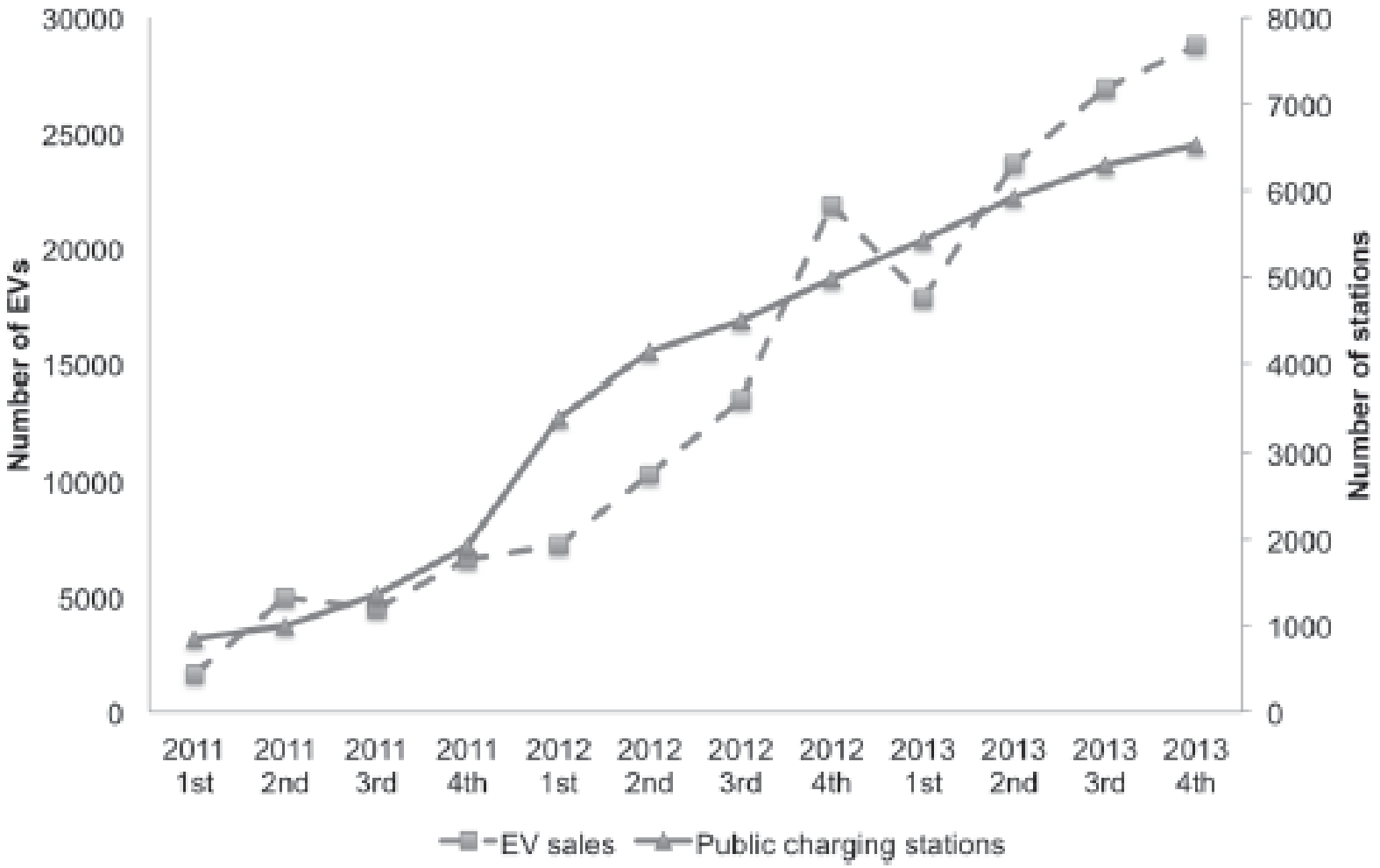
Article for today: Li et al. (2017)

Outline

Li et al. (2017) study the presence of indirect network effects on the adoption of plug-in electric vehicles (EVs) and their policy implications:

- Look at the US divided into 353 Metropolitan Statistical Areas (MSA).
- Quarterly EV sales and charging station deployment from 2011 to 2013.
- Only look at the scale of the charging network but abstracts from e.g. compatibility and locational choices.

Main object of interest



Central results

- A 10% increase in no. charging stations increases EV sales by 8.4%.
- A 10% increase in the EV stock increases no. charging stations by 6.1%.
- Estimate that the federal income tax credit on EV purchases contributed to about 40% of EV sales.
- Of the 40%, 40% are explained by feedback effects from indirect network effects and not the direct effect from the subsidy.
- A policy of equal government spending targeted at subsidizing charging station investments would have been more than twice as effective in promoting EV adoption (at the low station cost).

Policy implications of indirect network effects

We distinguish between **normative** and **positive** policy implications:

- If the network effects lead to *externalities*, decentralized EV adoption is not necessarily at the socially optimal level.
- Li et al. (2017) says nothing about optimal EV adoption.
- Comments on the most *cost-effective* policy where countries already have a target for EV adoption:
  - The policy target is taken as given.
  - Provide answers to under what conditions, different policies are cost-effective in reaching such a target.

Article for today: Li et al. (2017)

Microfounding the supply side

Setup

- In period t, the network consists of N_t firms each operating one station (i.e. N_t is also the number of charging stations).
- Each firm is infinitely lived and maximizes the stream of per period profits, discounted with the factor R.
- A station firm, which enters the market in period t, has to incur the fixed costs F_t of investing in and installing a charging station.
- From period t, the firm can generate per period profits π_{t+s} for $s \geq 0$.
- On the demand side each consumer holds at most one EV at any given time.
- The total number of users in the network is Q_t, which is the total stock of EVs in the economy at time t.

The short run: profit per station

Stations have constant marginal costs c_t and symmetric demand, and set prices each period. In the symmetric Nash Equilibrium:

- Prices are a markup over marginal costs, where the markup falls with the number of stations:

$$p_t^* = c_t + \varphi(N_t), \quad \varphi(N_t) > 0, \quad \varphi'(N_t) < 0$$

- $D_t(p_t^*)$ is an EV owner's total charging demand. Stations share the market equally, so each sells $Q_t D_t(p_t^*)/N_t$.
- Each firm therefore receives per-period profits

$$\pi_t^* = \frac{Q_t D_t(p_t^*)}{N_t}\varphi(N_t) = Q_t f(N_t), \quad f(N_t) \equiv D_t(p_t^*)\varphi(N_t)/N_t. \quad (1)$$

Derivation: section 2 of the note on Li et al. (2017).

Why does the network grow with the fleet?

Take (1) as given: $\pi_t^* = Q_t f(N_t)$. A firm can enter now or wait.

1. The present value of entering in t and in $t+1$, and the condition under which a firm is indifferent.
2. The equation for $\ln(N_t)$, and γ_1 in terms of the elasticity λ_1 of f.
3. The sign λ_1 must have for $\gamma_1 > 0$, and what it means.

Stop point

The entry condition, γ_1 in terms of λ_1, and the sign of λ_1.

The full questions and the stretch are on the worksheet.

Solutions 1, Q1–Q3: the questions

Q1. When is a firm indifferent between entering now and next period?

Q2. What is the equation for $\ln(N_t)$, and what is γ_1?

Q3. Which sign must λ_1 have for $\gamma_1 > 0$, and why?

Article for today: Li et al. (2017)

The demand side

EV regression equation

Demand comes from a logit model of discrete choice, where mean utility is linear in $\ln(N)$. The preferred **estimating** EV equation is:[1]

$$\ln(q_{k,m,t}) = \beta_0 + \beta_1 \ln(N_{m,t}) + \beta_2' X_{k,m,t} + T_t + \delta_{k,m} + \epsilon_{k,m,t}$$

⏩ The logit model

[1] The logit implies $\ln\left(s_{k,m,t}\right) - \ln(s_{0,m,t})$ on the left-hand side. The two specifications give almost identical estimates (Li et al., 2017, footnote 17 and table 6).

Article for today: Li et al. (2017)

Transition dynamics

A simplified model

To sum up, the two estimating equations are

$$\ln(q_{k,m,t}) = \beta_0 + \beta_1 \ln(N_{m,t}) + \beta_2' X_{k,m,t} + T_t + \delta_{k,m} + \epsilon_{k,m,t} \tag{2a}$$

$$\ln(N_{m,t}) = \gamma_0 + \gamma_1 \ln(Q_{m,t}) + \gamma_2' Z_{m,t} + T_t + \mu_m + \varsigma_{m,t}. \tag{2b}$$

To study the dynamics of the system in (2), we will consider a simplified model:

$$\ln(q_t) = \beta_1 \ln(N_t) + \beta_2 \ln(p_t) + \beta_3 x_t, \tag{3a}$$

$$\ln(N_t) = \gamma_1 \ln(Q_t) + \gamma_2 z_t \tag{3b}$$

$$Q_t = q_t + \delta Q_{t-1} \tag{3c}$$

Deriving a transition equation

From the transition equation in (3c), we note that

$$q_t = Q_t - \delta Q_{t-1} \tag{4}$$

Plugging (4) and the network equation in (3b) into the EV equation in (3a) and taking exponentials, we get

$$Q_t - \delta Q_{t-1} = Q_t^{\gamma_1 \beta_1} \exp\{c_t\}, \quad c_t \equiv \beta_1 \gamma_2 z_t + \beta_2 \ln(p_t) + \beta_3 x_t \tag{5}$$

In the following, we will assume that c_t is constant, i.e. $c_t = c$.

Steady State

In steady state (st.st.) all variables are constant and we therefore drop subscript t and let "*" indicate the st.st. value.
Solving for Q^* in (5), we see that we have two potential st.st. solutions:

$$Q^* - \delta Q^* = (Q^*)^{\gamma_1 \beta_1} \exp\{c\} \quad \Leftrightarrow$$

$$Q^* = 0 \ \vee \ Q^* = \left(\frac{\exp\{c\}}{1-\delta} \right)^{\frac{1}{1-\gamma_1 \beta_1}} > 0 \tag{6}$$

- $Q^* = 0$ is a *degenerate* equilibrium, where no one adopts an EV and no firm installs a charging station.
- This equilibrium is rather uninteresting because no path with a positive stock leads to it. We will focus on $Q^* > 0$.
- Feedback scales the long-run effect of a permanent price change by $1/(1 - \gamma_1 \beta_1)$.

Transition diagram

(5) defines Q_t as an implicit function of last period's stock, $Q_t = G(Q_{t-1})$. For $\gamma_1\beta_1 \in (0,1)$ and $\delta > 0$:

- G is increasing, $G'(Q_{t-1}) > 0$,
- returns are diminishing, $G''(Q_{t-1}) < 0$,
- the slope ends up below one as $Q_{t-1} \to \infty$.

Draw the transition diagram.

Proof: section 4 of the note on Li et al. (2017).

Do we have a critical mass here?

Q: Do we have a critical mass here?

Intuition?

Article for today: Li et al. (2017)

Cost-effective EV policies

Policy implications of network effects

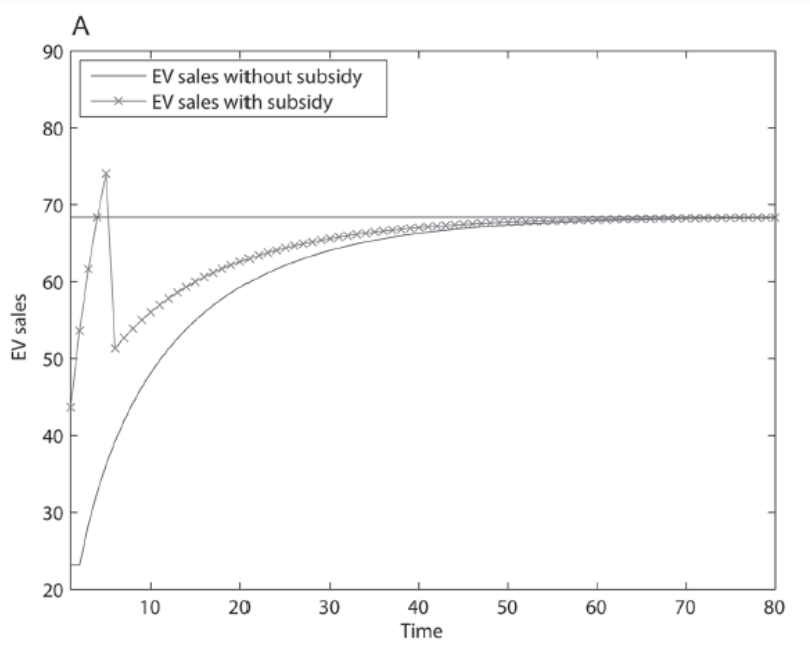
To gain intuition of the policy implications of indirect network effects, Li et al. (2017) conduct two policy experiments in the simplified model:

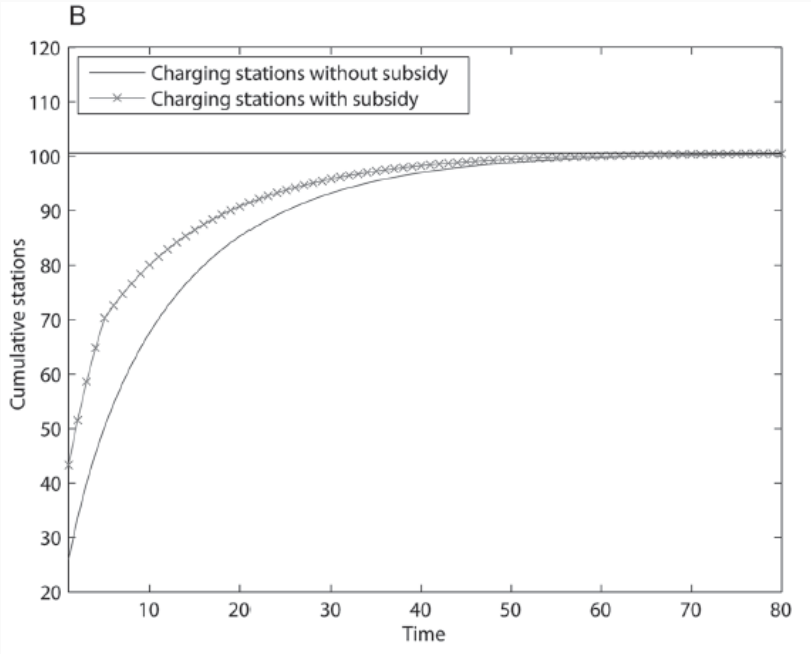
1. Provide a *temporary* subsidy to EV purchases of $7,500 for the first five periods.
2. Assume the expenditures from subsidizing EV purchases are used to *temporarily* subsidize charging stations.

Policy 1 is used to illustrate the presence of feedback effects.

A comparison of Policy 1 and Policy 2 is used to derive the most *cost-effective* policy.

Example 1: Feedback effects





Source: Li et al. (2017)

- The *temporary* subsidies do not change the long-run st.st. solution.
- However, subsidy speeds up the adoption of EVs.
- The effect of the subsidy lasts longer than its implementation period due to feedback effects.

Example 1: Feedback effects

Understanding feedback effects in terms of cross-price elasticities:

Table 4: Sample of Mean Own- and Cross-Price Elasticities for EV Models

AEV Make and Model	i3	Soul	i-Miev	Leaf	E-Golf	E-Up!
No Feedback Effects						
BMW i3	-2.0456	0.0016	0.0015	0.0015	0.0017	0.0016
Kia Soul	0.0020	-1.6572	0.0018	0.0019	0.0020	0.0019
Mitsubishi i-Miev	0.0002	0.0002	-1.6530	0.0004	0.0002	0.0002
Nissan Leaf	0.0031	0.0030	0.0026	-1.8759	0.0044	0.0044
Volkswagen E-Golf	0.0082	0.0078	0.0072	0.0047	-2.0333	0.0047
Volkswagen E-Up!	0.0010	0.0009	0.0009	0.0011	0.0015	-1.5387
With Feedback Effects						
BMW i3	-2.0475	-0.0003	-0.0004	-0.0003	-0.0002	-0.0003
Kia Soul	-0.0007	-1.6600	-0.0009	-0.0009	-0.0007	-0.0009
Mitsubishi i-Miev	-0.0001	-0.0001	-1.6681	-0.0035	-0.0001	-0.0001
Nissan Leaf	-0.0009	-0.0010	-0.0149	-1.8936	-0.0040	-0.0087
Volkswagen E-Golf	-0.0033	-0.0037	-0.0042	-0.0036	-2.0416	-0.0036
Volkswagen E-Up!	-0.0001	-0.0001	-0.0002	-0.0014	-0.0012	-1.5411

Notes: The top panel of the table reports the mean price elasticities of AEV models without accounting for feedback effects, while the bottom panel shows them accounting for feedback effects. Each cell entry, where i denotes rows and j denotes columns, provides the percentage change in market share of model j with respect to a 1% change in the price of model i.

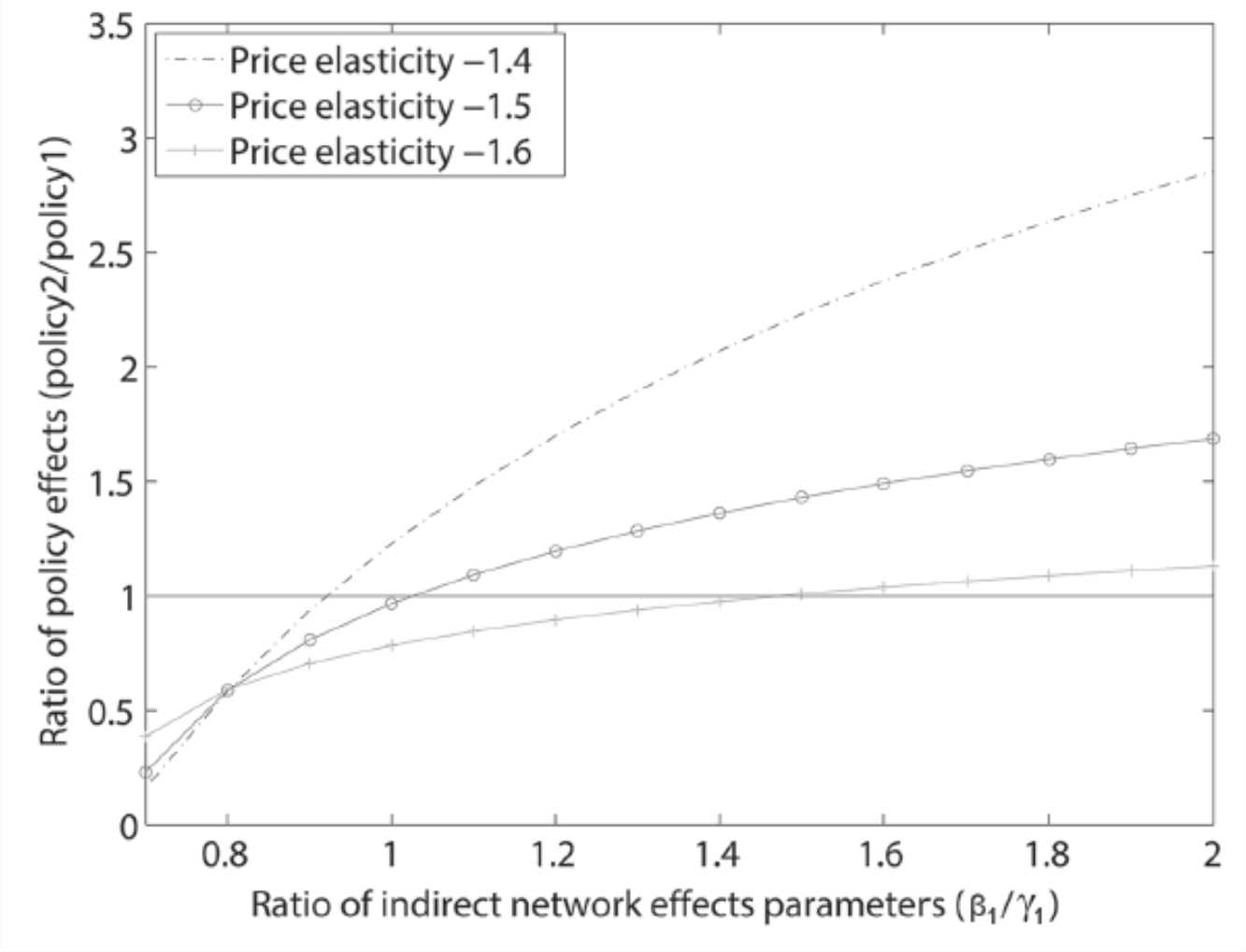
Source: Springel (2021)

Example 2: Cost-effectiveness of Policy 1 vs. Policy 2

- In many countries, emissions from transport are not taxed.
- Instead, the policy maker can subsidize EVs.
- Assume the policy maker has financed $ Y of expenditures to subsidize the adoption of EVs.
- We assume that the policy maker is interested in as many additional EV sales as possible.
- Should the policy maker use Policy 1 or Policy 2?

The cost-effectiveness is measured as the ratio of the two policies' cumulative sales increases, discounted at 5% per year.

Example 2: Cost-effectiveness of Policy 1 vs. Policy 2



Source: Li et al. (2017)

Example 2: Cost-effectiveness of Policy 1 vs. Policy 2

As the relative strength of the indirect network effect on EV purchases increases, it becomes more likely that subsidizing charging stations is the most cost-effective:

- When EV purchases react relatively more to charging stations than vice versa, then subsidizing charging stations creates the largest effect on cumulative sales.

For higher own-price elasticities of EVs, it is more likely that subsidizing EV purchases is cost-effective:

- If EV demand is relatively price elastic, EV subsidies stimulate EV demand relatively more.

Article for today: Li et al. (2017)

Estimation and identification

Estimation

The estimating equations are

$$\ln(q_{k,m,t}) = \beta_0 + \beta_1 \ln(N_{m,t}) + \beta_2' X_{k,m,t} + T_t + \delta_{k,m} + \epsilon_{k,m,t}, \quad (7)$$

$$\ln(N_{m,t}) = \gamma_0 + \gamma_1 \ln(Q_{m,t}) + \gamma_2' Z_{m,t} + T_t + \mu_m + \varsigma_{m,t}, \quad (8)$$

Simultaneity bias/Reverse causality: $N_{m,t}$ is endogenous in (7) and $Q_{m,t}$ is endogenous in (8).

Solution: Instrumental Variable (IV) regression.

Which way does simultaneity bias OLS?

Not in the paper. A stochastic version of the two simple equations, with $\delta = 0$ so that $Q = q$:

$$\ln(q) = \beta \ln(N) + \epsilon,$$
$$\ln(N) = \gamma \ln(q) + u,$$

where $\beta, \gamma > 0$ play the roles of β_1, γ_1, $\beta\gamma < 1$, and ϵ and u are uncorrelated with mean zero.

1. Trace a positive demand shock through the two equations.
2. $\ln(q)$ and $\ln(N)$ as functions of the two shocks.
3. The sign of $\text{Cov}(\ln(N), \epsilon)$, and the bias of OLS.

Stop point

$\ln(N)$ in terms of the shocks, the sign of the covariance, the direction of the bias.

Solutions 2, Q1–Q3: the questions

Q1. Trace a positive demand shock, $\epsilon > 0$, through the two equations.

Q2. Write $\ln(N)$ as a function of the two shocks.

Q3. What is the sign of $\text{Cov}(\ln(N), \epsilon)$? Does OLS over- or underestimate β?

IV estimation: A recap

An instrument should be

- Relevant: Should predict the endogenous explanatory variable.
- Valid: Should be uncorrelated with the error term and only affect the dependent variable through the explanatory variable.

Discussion: Discuss the two requirements and what you believe could be good instruments.

Instrument for charging stations

As instrument for $N_{m,t}$, they use
$\ln(\text{no. of grocery stores}) \times \ln(\text{lagged national stations})$:

- Why the interaction between the two variables?
- Relevance: Well, we can test it, but what is the intuition?
- Validity?

Estimation of EV equation

Table 3. EV Demand Equation

Variable	OLS (a)	OLS (b)	OLS (c)	OLS (d)	OLS (e)	GMM (f)
ln(no. of charging stations)	.378***	.389***	.502***	.352***	.177***	.844***
	(.017)	(.017)	(.021)	(.037)	(.036)	(.162)
ln(gasoline price) × PHEV	1.095***	1.186***	1.922***	.095	−.033	−.083
	(.203)	(.301)	(.353)	(.208)	(.201)	(.206)
ln(gasoline price) × BEV	.611***	.704***	1.908***	.560**	.386*	.419*
	(.203)	(.300)	(.406)	(.229)	(.217)	(.220)
ln(retail price − tax incentives)	−.480***	−.470***	−.817***	−1.378***	−1.433***	−1.288***
	(.034)	(.034)	(.179)	(.142)	(.140)	(.131)
ln(residential charging from EV project)	.046*	.035	.019	.026**	.059***	.050***
	(.025)	(.024)	(.022)	(.012)	(.011)	(.010)
ln(personal income)	.667***	.664***	.729***	2.018**	1.215*	2.056**
	(.162)	(.161)	(.179)	(.852)	(.695)	(.855)
ln(hybrid vehicle sales in 2007) × time trend					.031***	.011**
					(.003)	(.005)
ln(personal income) × time trend					.003	.001
					(.016)	(.020)
Year-quarter fixed effects	No	Yes	Yes	Yes	Yes	Yes
Vehicle model fixed effects	No	No	Yes	Yes	Yes	Yes
MSA-model fixed effects	No	No	No	Yes	Yes	Yes

Note. The number of observations is 14,563. The dependent variable is ln(EV sales) by model by quarter in an MSA. Retail prices are manufacturers' suggested retail prices. Tax incentives include both federal and state-level tax credits and tax rebates associated with EVs. All standard errors are clustered at the MSA level. ln(residential charging from the EV project) is the logarithm of quarterly number of residential charging stations that are built in an MSA by the EV Project, which is managed by ECOtality North America and supported by the US Department of Energy to deploy level 2 residential and commercial charging stations in 22 US cities. ln(hybrid vehicle sales in 2007) × time trend and ln (personal income) × time trend are included to address the concern that the charging stations trend could be correlated with potential trend of EV demand. The *p*-value for underidentification test for the last column is 0.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

GMM (f) is above every OLS column.

OLS is below IV in the EV equation. Why?

In Table 3, OLS is *below* GMM. Compare with your answer to part 2, Q3.

Q: What must be true of the two shocks, ϵ and u, for OLS to fall below the true β? What could such shocks be?

Instruments for vehicle stock

As instruments for $Q_{m,t}$, they use current gasoline price as well as three lagged terms of the gasoline price:

- Relevance: Well, we can test it, but what is the intuition?
- Validity? (In this case, we can actually test it, why?)

Estimation of charging station equation

Table 7. Charging Station Equation

Variable	OLS (a)	OLS (b)	OLS (c)	OLS (d)	GMM (e)	GMM (f)
ln(EV installed base)	.374***	.540***	.136***	.115***	.613***	.659***
	(.025)	(.028)	(.029)	(.028)	(.157)	(.157)
Charging station tax credit (%)	−.005***	−.003*	.003	.003	.012	.012
	(.002)	(.002)	(.013)	(.014)	(.014)	(.014)
Public funding or grants	.099*	.077	.007	−.020	.078	.088
	(.060)	(.057)	(.048)	(.047)	(.054)	(.055)
ln(no. of grocery stores) × ln(lagged national stations)	.042***	.030***	.183***	.118***	.063**	.058**
	(.005)	(.005)	(.017)	(.020)	(.025)	(.025)
ln(hybrid vehicle sales in 2007) × time trend				.018***	.009**	.009*
				(.003)	(.004)	(.005)
Year-quarter fixed effects	No	Yes	Yes	Yes	Yes	Yes
MSA fixed effects	No	No	Yes	Yes	Yes	Yes
Overidentification test (*p*-value)					.3435	.1347
Underidentification test (*p*-value)					.0000	.0001

Note. The number of observations is 4,236. The dependent variable is ln(total number of publicly accessible charging stations) in an MSA in a quarter. Tax credit for charging stations is the state-level tax credit to cover the cost of installing charging stations and is measured in percentage. Public funding is a dummy variable indicating whether public grants for building charging stations are available. ln(no. of grocery stores) × ln(lagged national stations) is the instrumental variable used in the EV demand equation. Column *e* is the preferred specification with gasoline prices in previous years as the IV for ln(EV installed base). Column *f* adds tax incentives for EV purchases as an additional IV. All standard errors are clustered at the MSA level.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

GMM (e) is above every OLS column.

OLS is below IV in the station equation. Why?

In Table 7, OLS is again *below* GMM. The stretch in part 2 gives the direction of the simultaneity bias in $\hat{\gamma}$.

Q: What must be true of ϵ and u for OLS to fall below the true γ? What could such shocks be?

Article for today: Li et al. (2017)

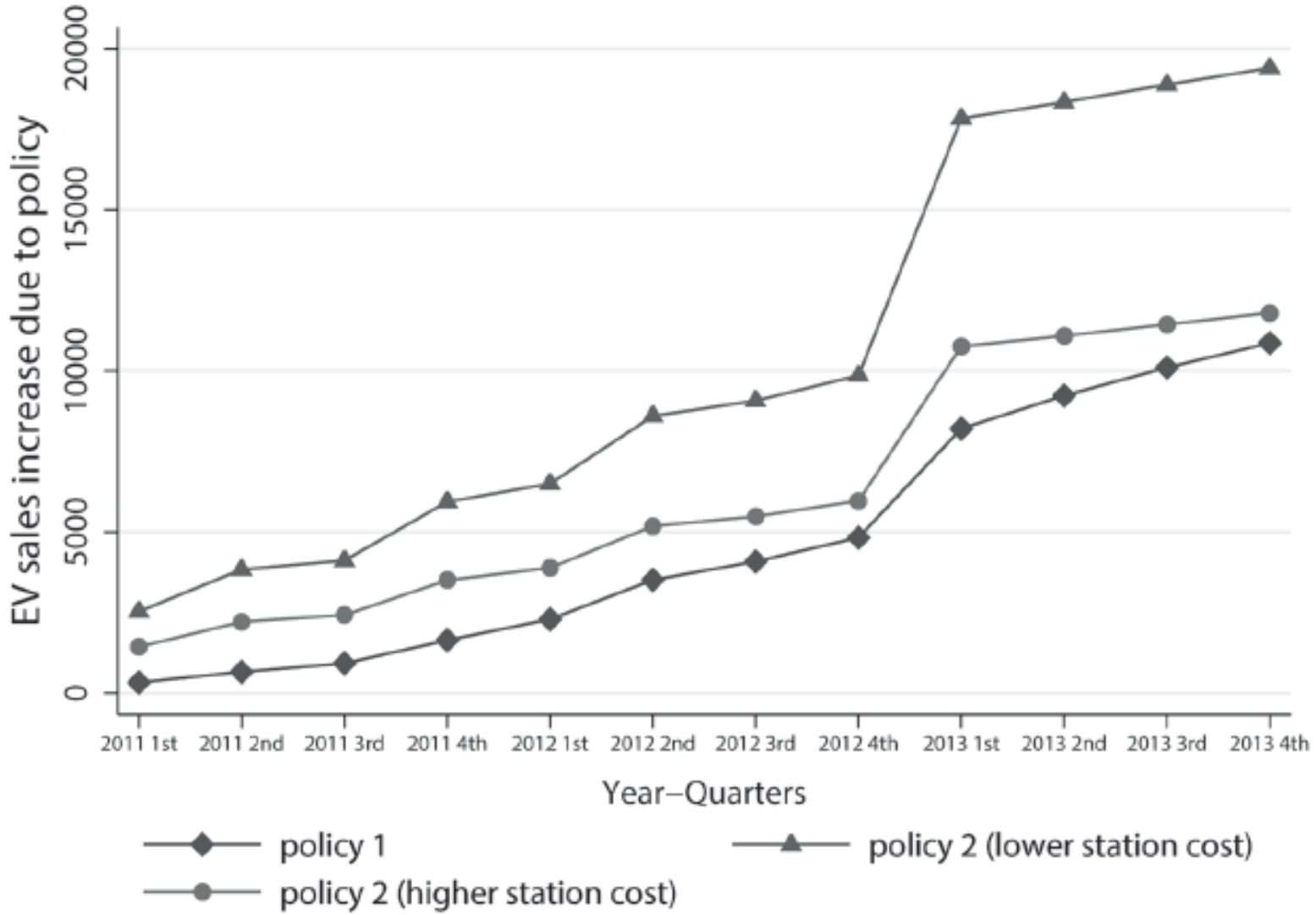
Counterfactual scenario

Counterfactual experiment

- EVs purchased in or after 2010 are eligible for a federal income tax credit up to $7,500.
- The credit expires once 200,000 qualified EVs have been sold by each manufacturer.
- Between 2011-2013, a total of $924.2 million was given in tax credits.

- **Policy 1:** simulate EV sales without the tax credit. The gap to actual sales is the effect of the credit.
- **Policy 2:** give the same $924.2 million to charging stations instead, and compare cumulative EV sales with Policy 1.

Cost-effectiveness of Policy 1 vs. Policy 2



Significant heterogeneity in cost-effectiveness

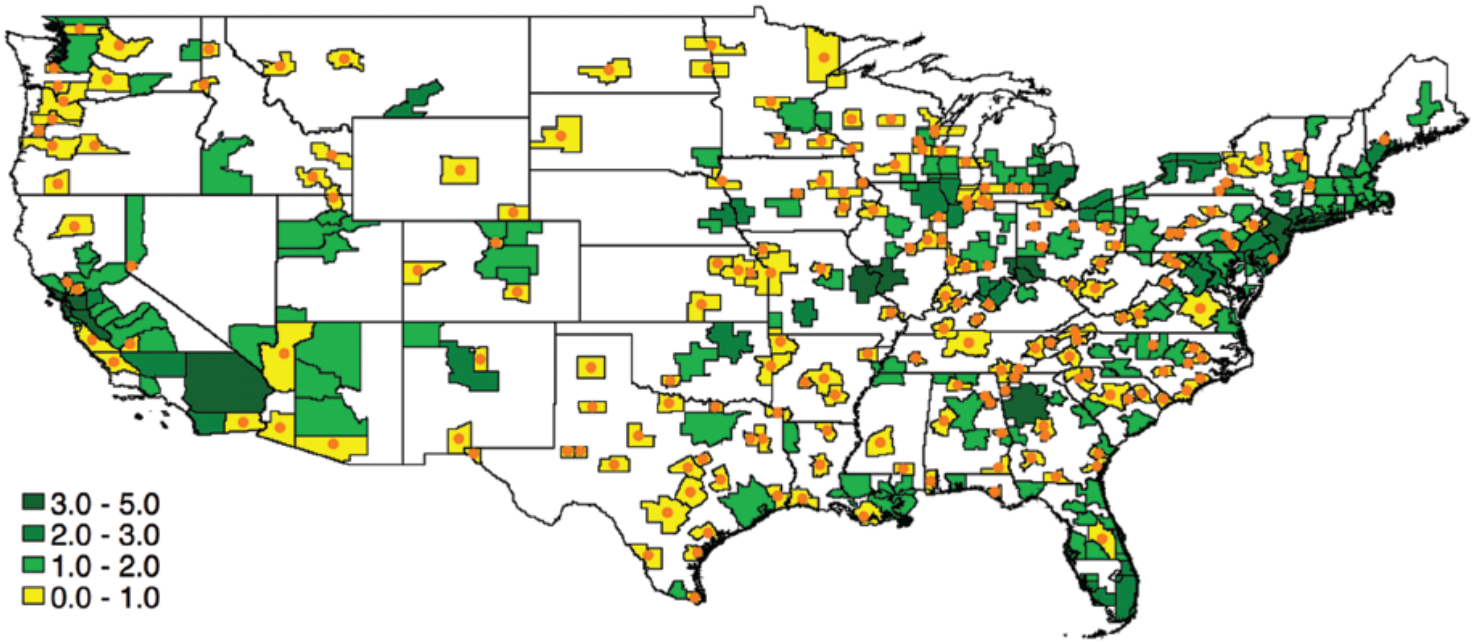


Figure 7. Heterogeneous policy effectiveness (policy2/policy1). Policy 1 gives new EV buyers a tax credit of $2,500–$7,500 based on different models as the current income tax credit policy for EVs. Policy 2 builds charging stations in all MSAs with the same budgetary cost as policy 1, assuming the investment cost per station being $27,000. The figure plots the ratio of the EV increases due to the two subsidy policies. The policy effectiveness of policy 2 varies across locations due to the heterogeneous impacts of public charging stations on the EV demand. The regions with the dots are locations where policy 1 is more effective.

Rule of thumb: push the charging network early

Early in the transition, money on stations buys more EVs than money on purchases:

- Li et al. (2017), US 2011–2013: early adopters care little about the price, so a purchase subsidy mostly pays people who would buy anyway. EV demand responds strongly to stations, so one more station adds much to the value of an EV.
- Springel (2021), Norway 2010–2015: a dollar on station subsidies gave more than twice as many extra EVs as a dollar on price subsidies. With more spending the ranking flips, because the effect of stations tapers off faster.

Conclusion

Key take-aways

1. Cars and chargers reinforce each other: the elasticities are 0.84 and 0.61 in Li et al. (2017).
2. Feedback amplifies policy: the long-run effect of a permanent price change is scaled by $1/(1-\gamma_1\beta_1)$, and a temporary subsidy works after it has ended.
3. With $\gamma_1\beta_1 < 1$ at all market sizes there is no critical mass, and a temporary subsidy speeds up adoption without moving the long run. A critical mass needs an elasticity above one in the thin market.
4. Simultaneity alone makes OLS overestimate both elasticities. In Li et al. (2017) OLS is below IV. In our model with $\delta = 0$ that needs strongly negatively correlated shocks, less so with a stock.
5. At the low station cost, the same money on stations would have been more than twice as effective in the US in 2011–2013. The ranking depends on the relative strength of the two network effects and on how price-sensitive buyers are.

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Appendices

The demand side in one slide

Vehicle adoption is typically modeled as a *discrete choice*: the consumer chooses the alternative k with the highest utility $U_{i,k} = V_k + \eta_{i,k}$, where $k = 0$ is not buying.

With IID extreme value errors, the choice probabilities are the logit formula. Normalizing $V_0 = 0$, log odds equal mean utility, and shares s replace probabilities:

$$\mathbb{P}_k = \frac{\exp\{V_k\}}{\sum_{j=0}^{K} \exp\{V_j\}} \quad \Rightarrow \quad \ln(s_k) - \ln(s_0) = V_k$$

Mean utility is assumed to be linear in $\ln(N)$, so the number of stations has a diminishing effect.

Derivation: section 3 of the note on Li et al. (2017).

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